

NSS 69th Round Report

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Introduction to the NSSO

The National Sample Survey is conducted by the Ministry of Statistics and Programme Implementation. The survey is conducted annually over a period of 6 months to a year. As part of its 69th round, the National Sample Survey Organisation conducted a nationwide survey on Drinking Water, Sanitation, Hygiene and Housing conditions in India in 2012. In addition they also conducted a survey on 'Particulars of slums'. The objective of the survey was to collect information on the different aspects of living conditions necessary for decent and healthy living of the household members and develop suitable indicators (NSS report).

Understanding the Data

Looking at the summary statistics of the NSS data will help in interpreting the results later and provide useful context. They also let us glean into the type of data being analyzed and they help inspire thoughtful research questions.

Social Group	Frequency	Percent	Cumulative
ST	8,614.1348	9.02	9
SC	17,835.841	18.67	27
OBC	40,996.929	42.91	70
Other	28,101.095	29.41	100

Area type where dwelling is located	Frequency	Percentage	Cumulative Share
Notified Slum	2,156.6369	2.26	2.26
Non-Notified Slum	1,116.23988	1.17	3.43
Squatter Settlement	65.1567578	0.07	3.49
Other Areas	92,178.966	96.51	100.00

We look at caste and dwelling demographics as these are strong driving factors for many of the indicators of freedom that will be analyzed subsequently. The data shows that the largest caste group in India is the OBC making almost 40% of the population and cumulatively making up 70% of the population with SC and ST combined. We see that although a vast majority of people do not live in slums or as squatters they make up a nontrivial aggregate of around 5.8% of the population, that is roughly more than 82 million people.

Our Research Idea

The NSSO surveyed a sample of over 95000 households. 5.6% of the households surveyed were living in slums. We wanted to understand the living conditions of the 78 million people making up India's poorest. India's history is full of stories of the oppressors and the oppressed. The historic injustices towards the "lower" castes and the grave inequalities that have plagued the nation from the time of its conception and long before still continue to impact them today. The state has been investing continuously to improve the capabilities of these peoples and end the violence that they have faced for generations. It therefore becomes important for us to quantify the changes in the freedoms of these peoples by looking at fundamental rights. Here we attempt to use this NSS data to look at the social demographic of the poorest in this country and their living conditions as an indicator of their freedoms.

Before we start with the results, a cursory overview of the data and familiarization with some frequently used terms would be prudent:

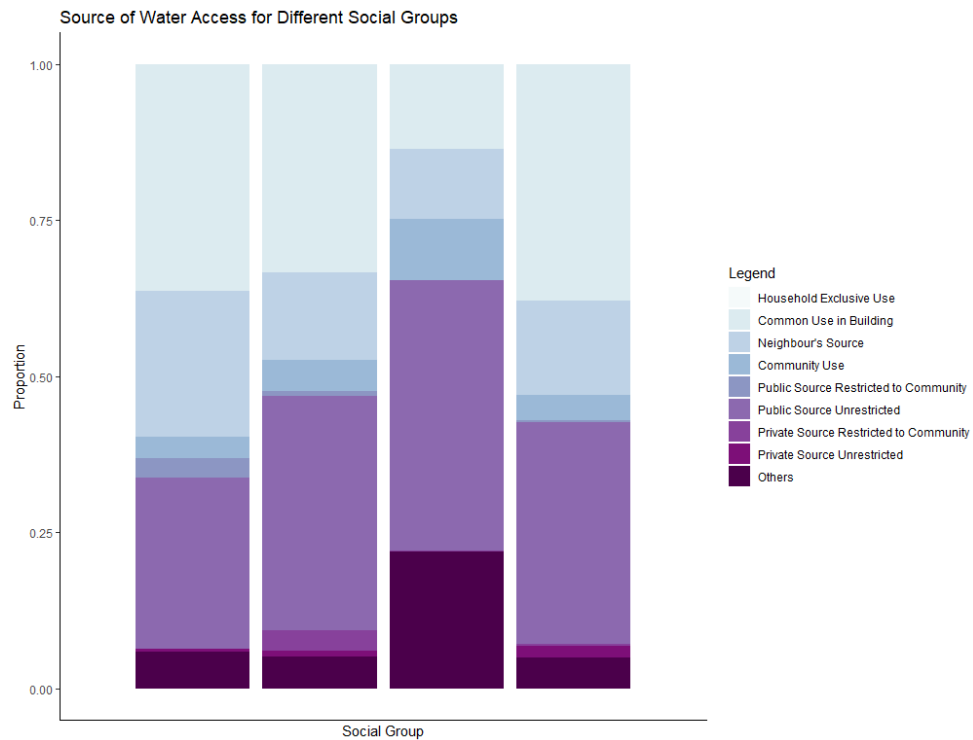
1. Notified slums: These are areas notified as slums by the concerned State governments, municipalities, corporations, local bodies or development authorities.
2. Non-notified slums: In this survey any compact settlement with a collection of poorly built tenements, mostly of temporary nature, crowded together, usually with inadequate sanitary and drinking water facilities in unhygienic conditions, is considered a slum provided at least 20 households live there. If such a settlement is not notified as a slum, it will be called a non-notified slum.
3. Slums: The word "slum" refers to both notified slums and non-notified slums.
4. Squatter settlement: Slum-like settlements with less than 20 households are considered to be squatter settlements.

Data Analysis

We started our data analysis by looking at social groups.

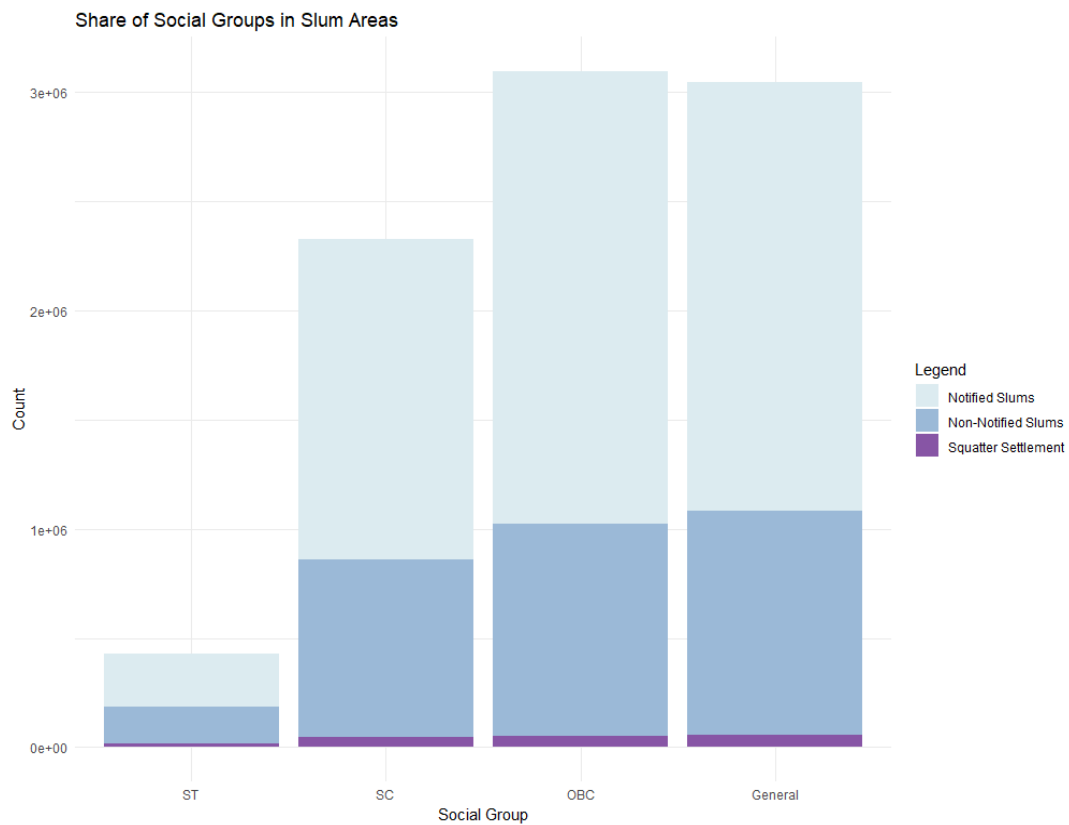
1. Understanding Water Access for Different Social Groups

By a considerable margin, we see that sources of water which we would associate with higher status are accessed by a higher proportion of upper castes in relation to that proportion of SC, ST, and OBC that have access to the same. Those same groups as such also show a greater reliance on common sources of water, as seen by the graph below.



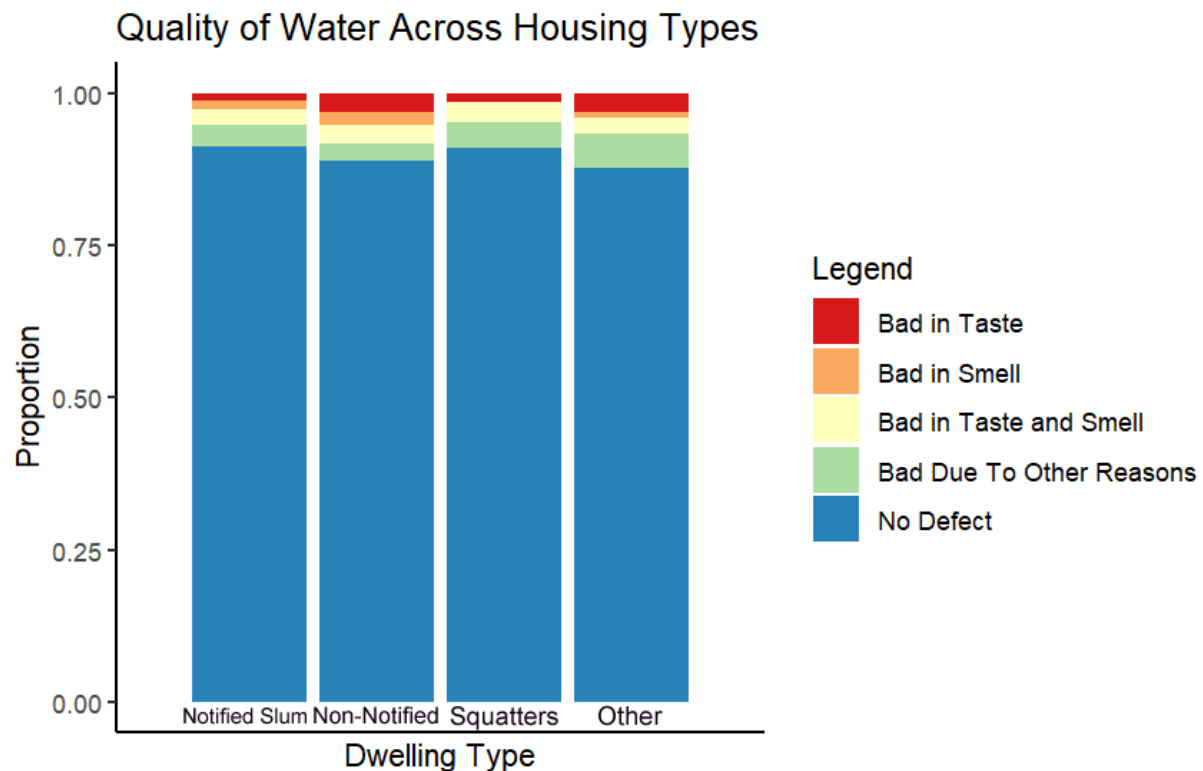
This can be viewed through the capabilities framework for freedom wherein the privilege to have exclusive access to water is viewed as an enhancement of one's capabilities.

2. Understanding share of social groups in slums

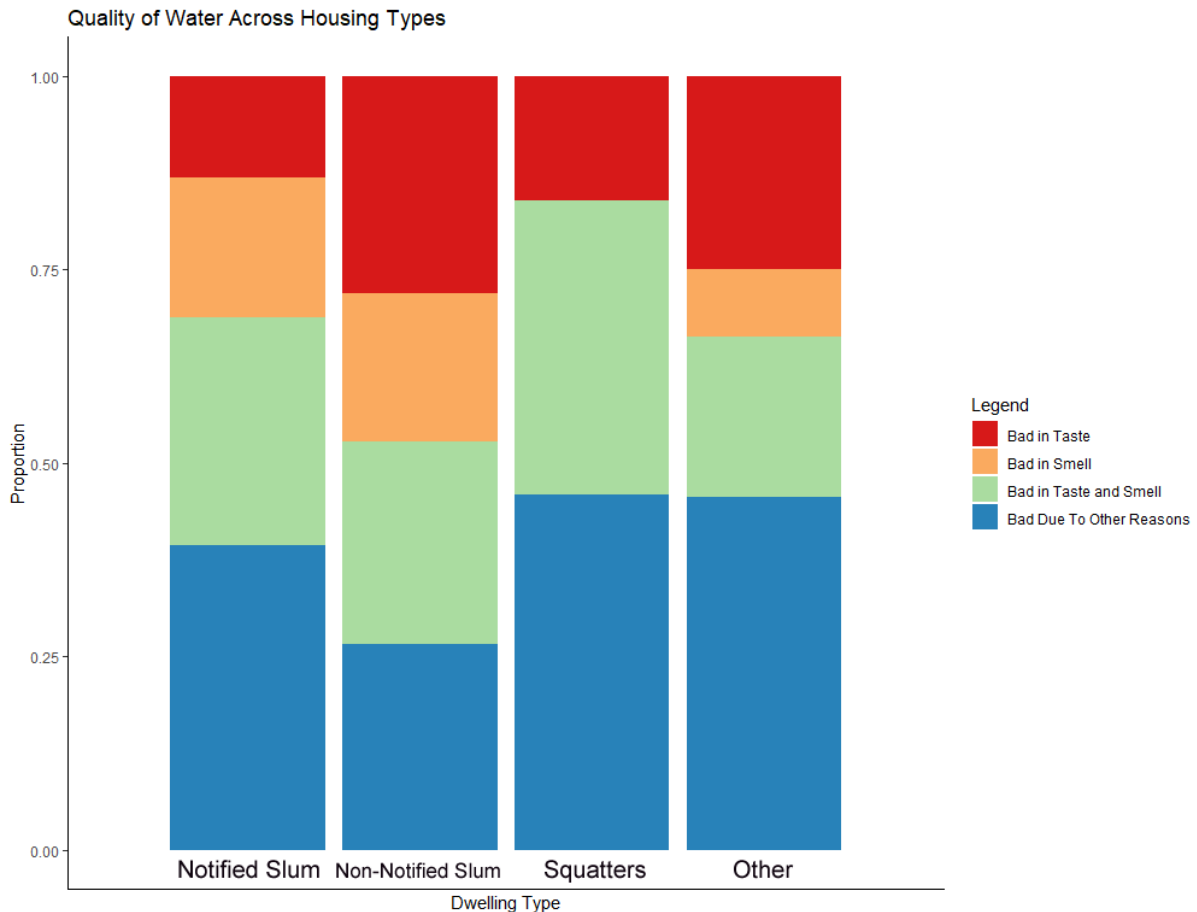


Across India, the largest social group residing in slums are the people belonging to other backward castes. Within every social group, more people live in notified slums. This is not surprising as the proportion of households in notified slums is 2.26% which is higher than the proportion of those in non notified slums (1.17%) and squatter settlements (0.07%). It is also seen from this that the number of ST, SC and OBC people living in slums is much larger than that of the general category, which indicates that people in the general category face less deprivation in terms of living conditions as compared to socially disadvantaged groups.

3. Understanding the quality of water across housing types



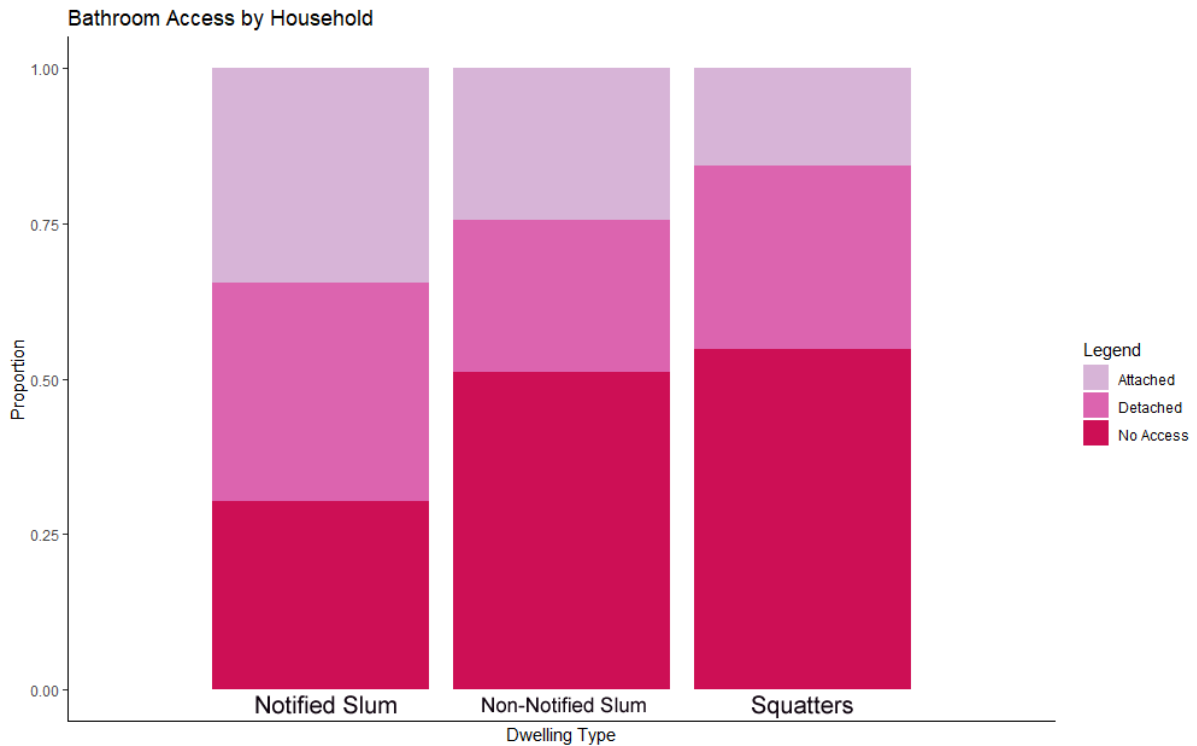
Access to safe drinking water has direct implications to the rights of an individual. 87.8% of India has access to good drinking water. It is clear that in addition to having no stable housing people in squatter settlements are also the group of people most affected by sub par water quality. Using the above graph for any further interpretations will be loose at best; hence, we removed the data points with a satisfactory water quality so as to better interpret water conditions of those across dwelling types.



The normalized graph above only contains the proportions of sub par water quality categorized by taste, smell, both taste and smell and other quality issues. Here it is difficult to make any remarks on which water quality is worse (e.g. is bad-smelling water worse than bad-tasting water?). A more detailed set of questions on the precise reason the water is bad would have helped in accessing such an important parameter of freedom as access to safe water.

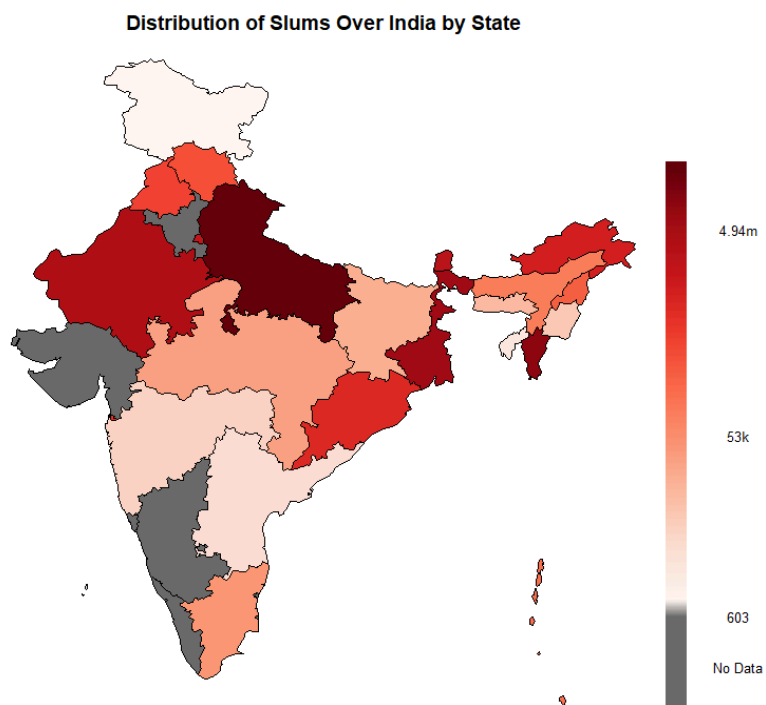
4. Does it matter whether a slum is notified?

The question then arises are there external factors that could have affected the answers. Does it just matter whether the household is in a slum area or not? One possible external factor could be whether the slum is notified.



If the slum is a non notified slum, there is a higher chance of it having no bathroom which is an indicator of sanitation, and a proxy for measuring deprivation.

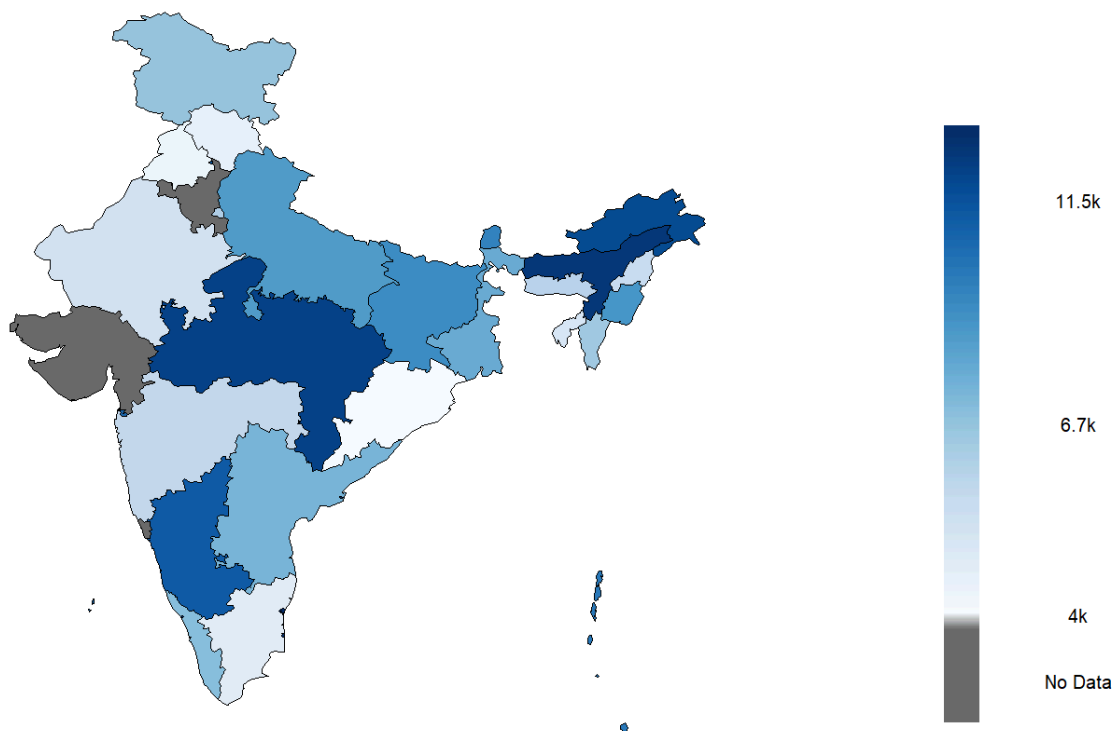
Another factor could be the location of the slum. The dataset used does not have enough information to answer these questions. However, we did try to understand what the distribution of slum households was statewise.



As seen in the map above, the distribution of slum households across India tends to be the most in the northern regions, with Uttar Pradesh and Uttarakhand having the highest number (4.9M), followed by Mizoram (650k), West Bengal (567k), and Rajasthan (340k). The lowest number of slum households, on the other hand, were present in Jammu & Kashmir (603), followed by Lakshadweep (1477), Tripura (1606), and Andhra Pradesh & Telangana (3623). There was no data present on slum households for Karnataka, Kerala, Gujarat, Goa, Haryana, and Puducherry.

Looking at this map on its own does not tell us much about the state of slum households in India, which is why we additionally tried to correlate this with consumption expenditure data across India:

Consumption Expenditure (in Rs.) Comparison Across India



As seen above, the highest consumption expenditures (on average, by state) were seen in Puducherry (11.5k) followed by Assam (11.1k), Madhya Pradesh (10k), and Arunachal Pradesh (10k). The lowest expenditures were seen in Odisha (4k) followed by Punjab (4.3k), Himachal Pradesh (4.5k) and Tamil Nadu (4.7k). As per the definition of consumption expenditure (money spent on satisfying *needs* and *wants*) we make the assumption that a higher consumption expenditure correlates to having enough financial resources to sustain the household, whereas a lower consumption expenditure correlates to being financially restricted. With this assumption in mind, we make our analysis.

On comparing the two maps, we can possibly infer that people within states like Rajasthan and Odisha (with relatively high numbers of slum households *and* low consumption expenditure) face higher deprivation. The same may also be said for the reverse case for

states like Madhya Pradesh, which have a lower number of slum households and high expenditures. However, there lie several exceptions to this theory (e.g. Maharashtra, with low no. of slum households *and* low expenditure, and Arunachal Pradesh, with high no. of slum households *and* high expenditure). Thus, without a dataset/survey specifically targeting slums across India, it is impossible to make any inferences on the extent of deprivation of those living in slums.

Conclusions and Remarks

Given the data, it is not possible to make conclusive measurements of the deprivation present across India vis-à-vis slums. There is clear evidence that households across India face severe issues with sanitation, and that regardless of social status (SC/ST/OBC/general) a significant proportion of India's population lives in subpar conditions. However, with the lack of detail within the questionnaire itself, it is difficult to find relationships between data across blocks that aid a meaningful analysis.

For example, we had considered using the presence of electricity as another measure of economic/living conditions; upon looking at the dataset, though, we found that most of India does have electricity. *However*, there is no indication on how reliable this source of electricity is. Nothing is said about the frequency of power cuts, whether electricity is only present in specific time constraints (e.g. only for a few certain hours in a day) and so on, which are all very important parameters to compare 'fairness' and comfort across the population—across state, social group, and rural/urban living areas. This setback is only one of the many we faced while trying to wrangle the dataset in a way that would provide us with meaningful and descriptive results.

The major part of our analysis has been with regards to the living conditions of those in slums; however, the dataset only looks at the number of slum households in a given area; not the number of *slums* altogether, or the size/population density of the slum. For example, the data shows that Maharashtra has very few slum households as compared to the rest of the states—however, it is also common knowledge that Maharashtra is home to some of the largest slums in not just the country, but the world (see: Dharavi, which is home to one million people). The data collected, then, can be misleading; this is not necessarily a fault with the methodology itself (as the survey looked at individual households rather than localities and focused on the general population rather than just slum dwellers) but having performed our analysis, we recognise that a more elaborate survey on socioeconomic conditions of those living in rural areas and slums needs to be conducted for us to draw conclusions on the extent of deprivation in India.

Contributions:

Anjali: Understanding the data, plots for point 1, 2 and their analysis, and Conclusion.

Maharshi: Introduction, plots for 3, 4, country plots and their interpretation.